Alternative Credit Scoring: Building Explainable AI for Financial Inclusion

MSIS 522 B — ADVANCED MACHINE LEARNING | HOMEWORK 1: THE COMPLETE DATA SCIENCE WORKFLOW

🚨 THE PROBLEM

📈 WHY ALT-CREDIT SCORING MATTERS

1.4B
UNBANKED ADULTS

45M
US "CREDIT INVISIBLE"

22%
DEFAULT RATE (TAIWAN)

🎯 PROJECT DEFINITION

Build an **end-to-end explainable credit scoring pipeline** demonstrating the complete data science workflow: from data acquisition through model comparison to interactive deployment with fairness auditing.

1. **Research & Data Acquisition**
   Identify, evaluate, and download 6+ credit datasets; document provenance and licensing
2. **Exploratory Data Analysis**
   Distributions, correlations, class balance, fairness subgroup breakdowns
3. **Predictive Modeling (5 Models)**
   Logistic Regression → Decision Tree → Random Forest → XGBoost → MLP Neural Network
4. **Explainability & Fairness**
   SHAP analysis (global + local), disparate impact, equalized odds across protected groups
5. **Interactive Deployment**
   5-tab Streamlit app with live predictions, SHAP waterfall, and ethics dashboard

✅ DESIRED OUTCOMES

Methodology: The Complete Data Science Workflow

DATASET RESEARCH → EDA → MODELING → EXPLAINABILITY → DEPLOYMENT

🔬 STEP 1: DATASET RESEARCH & SELECTION

Value: Rigorous dataset selection ensures data quality, licensing, and fairness-analysis fitness. 6 candidates evaluated.

DATASET	ROWS	COLS	SOURCE	TARGET	STATUS
⭐ **Taiwan Credit**	30,000	25	UCI	Default (0/1)	✓ DOWNLOADED
⭐ **German Credit**	1,000	21	UCI	Good/Bad	✓ DOWNLOADED
FICO HELOC	10,459	24	GitHub	Good/Bad	✓ DOWNLOADED
Give Me Some Credit	150,000	11	Kaggle	Distress (0/1)	CLI READY
Home Credit	307,511	122	Kaggle	Default (0/1)	CLI READY
LendingClub	2.9M	151	Kaggle	Charged Off	CLI READY

→

📊 STEP 2: EDA

Value: Uncover data quality, distributions, and class imbalance before modeling.

🤖 STEP 3: PREDICTIVE MODELING PIPELINE

Value: Compare 5 model families from baseline to deep learning. All use **70/30 stratified split** (`random_state=42`) + **5-fold StratifiedKFold** GridSearchCV.

🔍 STEP 4: SHAP EXPLAINABILITY

Value: Moves credit scoring from "black box" to **transparent, auditable**.

⚖️ STEP 5: FAIRNESS & ETHICS AUDIT

Value: Ensures AI decisions don't **discriminate by gender, age, or education**.

- **Disparate Impact:** DI ratio < 0.8 → flag (four-fifths rule)
- **Equalized Odds:** Compare TPR/FPR across SEX, AGE groups
- **Calibration:** Are probabilities accurate per subgroup?
- **SHAP Audit:** Protected features shouldn't dominate importance
- **Mitigate:** Reweight, remove, or threshold-adjust if biased

Findings, Outcomes & Impact

RESEARCH RESULTS → DEPLOYMENT DESIGN → AUDIENCE TAKEAWAYS

🏆 FINDING 1: DATASET SELECTION RESULTS

Primary Choice: Taiwan Credit Card Default

The optimal balance of **size (30K rows)**, **cleanliness (0 missing)**, and **fairness features (SEX, AGE, EDUCATION, MARRIAGE)**. Published academic benchmark with clear binary target.

30K
CLEAN ROWS

Why Taiwan Credit wins over all 6 candidates:

- ✅ **Right size** — Large enough for GridSearchCV + MLP, small enough for fast iteration
- ✅ **Zero missing values** — Skip imputation, focus on modeling
- ✅ **4 protected attributes** — Full fairness analysis possible (SEX, AGE, EDUCATION, MARRIAGE)
- ✅ **6-month payment history** — Rich temporal features for SHAP interpretation
- ✅ **22% default rate** — Natural imbalance motivates SMOTE, not so severe as to collapse metrics

📊 FINDING 2: EXPECTED MODEL PERFORMANCE

Based on published benchmarks on Taiwan Credit dataset:

XGBoost	~0.78 AUC
Random Forest	~0.77 AUC
MLP Neural Net	~0.76 AUC
Decision Tree	~0.72 AUC
Logistic Regression	~0.71 AUC

🖥️ OUTCOME: INTERACTIVE STREAMLIT APPLICATION

💡 WHAT THIS MEANS FOR THE AUDIENCE

The Bottom Line: Explainability ≠ Weakness

The best credit scoring model isn't just the most accurate — it's the one